

WIPIVIZ: A study on how WiFi Visualization can help users identify WiFi hotspots

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ABSTRACT

As the dependence on stable WiFi continues to grow, we understand that the placement of WiFi routers and devices requiring the signal will also become more important. WiFi signals can be affected widely by walls, doors, and nearby objects. However, the effect obstacles have on WiFi signal strength is hard to visualize and quantify for most consumers.

Our project, WIPIVIZ, aims to bridge this knowledge gap and help educate consumers about the ideal placement of routers to help them provide the most stable WiFi connection to devices in the room. Eventually, we would want our project, WIPIVIZ, to allow for a seamless way for users to use the VR headset to directly "place" and test virtual router locations in a room and visualize estimated WiFi hot spots, and figure out where the connection may not be as strong.

CCS CONCEPTS

• **Networks** → Network reliability; • **Human-Centered Computing** → Visualization.

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1 INTRODUCTION

Currently, many commercial and educational buildings can suffer from unideal placement of WiFi routers that may not be able to give the most optimal signal and connection to users. This problem is very prevalent in some of the classrooms and conference rooms that we see everyday on campus, sometimes even in Siebel. We believe that using WiFi visualization tools can be an asset in helping educate users on how WiFi signal strength can vary across the room and interact with various obstacles, eventually developing into a tool that can be utilized in deciding on router placement.

Many people do not understand the mechanics and subtleties of how WiFi signals actually interact with their surroundings. Concepts such as constructive and destructive interference, multipath

propagation, and attenuation through different materials are difficult to grasp for the average consumer. As a result, routers are often placed arbitrarily, producing dead zones and unreliable coverage. While existing educational text and images about these properties can help gain a baseline knowledge, being able to visualize these signals in an interactive way would provide an even better experience that can show the dynamic nature of these signals. With WIPIVIZ we set out to develop an experience that can help people do this using grid heat maps to show various WiFi signal "hotspots" where the signal is especially strong. We assessed the effectiveness of using such a visualization tool in helping teach the public about the nuances of wifi by performing a user study.

While prior research has introduced heatmaps, glyphs, spectrograms, and immersive VR systems for WiFi visualization, these papers have not focused on the educational aspect of using the tool. Heatmaps, which seem to be the most interpretable, are typically presented without evidence of how well participants interpret them. More complex methods such as glyphs or spectrograms demand technical expertise, and immersive systems like WaveRider rely on expert panels rather than broad user studies. This lack of empirical evaluation highlights a critical gap: we do not yet know whether these visualizations can actually help everyday users understand WiFi signal behavior.

Our work addresses this gap by conducting a user study that measures how effectively participants can use grid heatmaps to predict RSSI strength across multiple router placements, by testing their baseline knowledge, and then comparing it to their survey results after using the tool. We hope to gain evidence and insight on how effective these visualization tools are on helping WiFi non-experts predict signal strengths at various parts of the room. We originally wanted to do this user study comparing the baseline to grid heat map visualizations as well as glyph visualizations. Unfortunately due to time and resource restrictions, we were not able to fully finish updating the glyph visualization tool for the study, and had to conduct it using only the baseline and the grid heatmap.

There were a couple of challenges that we would face as we implemented our user study. We wanted to find a good way to measure how well the visualization tools would end up helping the user in understanding how the WiFi signal strength would vary across places in the room. We wanted metrics that can directly help us compare which tool was the most helpful and how much it helped. While it was suggested that we ask only which location

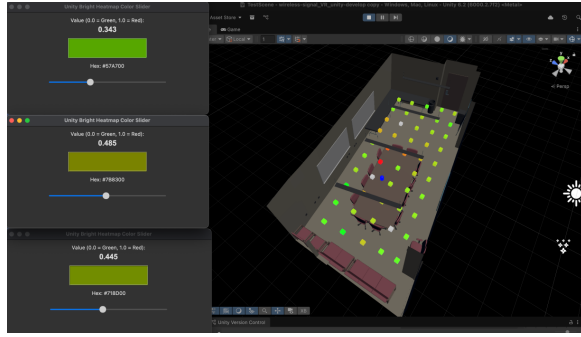


Figure 1: Survey example for router signal strengths

on the room would have the strongest signal from a set of points, we felt it would not be able to give us enough information on how much the visualization tools would have helped, as users may be able to guess purely based on distance away from the router. To circumvent this, we decided to associate signal strengths with a scale of colors ranging from red to green. We then asked the survey participants to guess the "strengths" of the signals by choosing a color, with red being the strongest, and green being the weakest. There were also a couple of other implementation challenges that we faced during the process, which will be talked about more later in the paper.

For the user study we developed our tool to create a grid heatmap that contained real RSSI signal values collected in a Siebel conference room using a laptop and raspberry pi, simulating the WiFi signal strengths with various "router" placements. Based on the signal strengths, we also mapped each of the 50 receiver locations to a color in a color scale from red to green. We also developed a slider system that survey respondents would use in order to estimate the receiver strength without having to guess the RSSI dbm value. We then surveyed respondents in the Illini Union with a less than expert level of knowledge in WiFi signal propagation. Overall, after we took the survey, we noticed that the results showed the heatmap did aid at least in part towards the understanding of WiFi signals in a room.

2 RELATED WORKS

Visualizing WiFi signals has been studied for decades. As such, there is a tall but narrow foundation upon which further study can be built upon, given there are few viable and commonly used methods available today.

Many original works began with heatmaps [2] [3], considering the ease with which data can be recorded and mapped. While heatmaps can contain valuable information in an easily understandable format, they can't convey how a signal can move through an environment alone, nor distinguish between multiple signal paths in one location.

This was the predominant method of visualization, until glyphs [6] were introduced. Glyphs enabled an easy way to understand visualization of a signal as it propagated through an environment, and could better distinguish between signals from multiple sources. The NCSA WiFi Visualization project upon which WIPIVIZ is based on uses a mixture of glyph visuals in the initial simulation run and a

Comparisons			
Approach	VR Capable	User-friendly	Signal Path Visual
Heatmaps [2] [3]	✓	✓	✗
Spectrograms [4]	✗	✗	✗
WaveRider [1]	✓	✗	✓
WIPIVIZ	✓	✓	✓

Table 1: WIPIVIZ vs. Existing Approaches

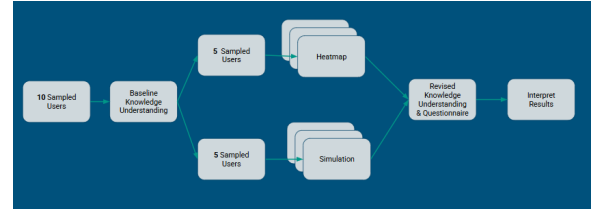


Figure 2: Survey Flowchart

resulting heatmap which communicates to the user where best to receive WiFi signals in the environment. This heatmap is modified from its source to display color to the user, so as to better show differences in RSSI value across grid locations.

Current research still explores the problem of WiFi visualization. While spectrograms and channel graphs have existed [4], only recently have the concepts been applied to WiFi visualization [7]. This is still an emerging field, but up to this point, these methods are difficult to parse by the layman, and are thus unsuitable for a more broadly available visual.

A unique system named WaveRider [1] uses a technique called Line Integral Convolution to visualize vector fields into streaks. These streaks are then layered and mapped onto a virtual environment created beforehand, making this VR capable. This system can also clearly distinguish between multiple signals thanks to the way the LIC contour lines show intermediate points. However, due to the use of vector fields and contour lines, the system visual may be confusing to those not adept in WiFi signal propagation. In fact, the authors mention their lack of a user study, instead recruiting a panel of experts, which brings bias to their answers.

A more general comparison between this paper's system and currently available approaches is displayed in Table 1.

3 SYSTEM OVERVIEW

As we create the user study survey, we wanted to make sure that we were collecting data in a way that would be easy for the survey participant to easily grasp, as well as give us some meaningful results on how the tools have helped the respondents. To do this we came up with the following plan as to how we wanted to run the survey shown in figure 2.

Since each person would be coming in with varying knowledge of WiFi signal propagation, we thought it would be important to get everyone's baseline score without the tools. We then show half the respondents the grid heat map and the other half the glyph

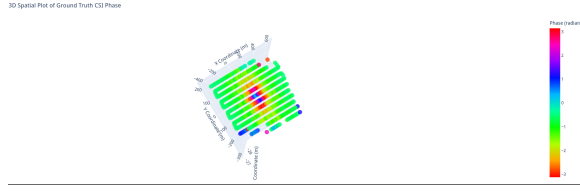


Figure 3: Simulation CSI

simulator with ray tracing. Our main analysis comes from comparing each group final survey score to their respective baseline score, and seeing which group improved. Unfortunately since we did not have the glyph simulator working due to technical challenges we performed our user study with only the grid heat map. Ideally we would have also liked to survey more people but due to time restrictions, this was not possible.

In the survey, we hid the colors of three different receivers, and wanted the users to estimate which color the hidden receivers would be and which would be the strongest. To get an idea of their baseline knowledge, we hid all of the colors, and had the respondents blindly guess, only telling them that the red color correlates to strong signal and green correlates to weaker signal. After getting the baseline for the participants, we let them play around and move within the simulator, and let them reference the surrounding grid heat map colors to make their new guesses for the the same hidden receivers.

4 PROPOSED APPROACH

The approach was to gather participants for the study and show them an initial image where there were three receivers that were colored gray. They would use the color slider mentioned initially to pick an interpolated value between 0 and 1, where 0 was the lowest recorded receiver value (green) and 1 was the highest recorded receiver value (red). The participants first pick a value without knowledge of the surrounding heatmap based on the location of the receiver relative to the blue transmission blob in the room. These initial guesses are recorded. From there, we will play the simulation and the other receivers will show up with their respective colors ranging from green to red as seen in figure 1. The three points of interest are still grayed out, and the participants will use the new receivers that create a heatmap to re-select the values of the grayed out receivers. These values are recorded in a google form we created, and the participants were asked to rank which gray receiver they thought had the strongest RSSI signal before and after the heatmap is shown.

We were also going to compare the real-world results we utilized for the user study compared to the simulations that we would create in Unity. However, the timeline for the project allowed us only to mimic a basic simulation of our room without incorporating Unity. The results of the simulation are shown in Figure 3. In the future, this simulation code can be used to simulate the room we used in figure 1 and can be compared to our measurements to see if the simulation gets similar results to the real-world.

5 IMPLEMENTATION

To make our approach possible, we needed a 3D model of a room which we could physically enter to record data for our study. This was provided to us by the NCSA project's creators as an fbx file of Siebel room 1218.

For our heatmap data collection, we used a Raspberry Pi 4, with an integrated WiFi transmitter, in conjunction with a USB WiFi receiver. The Raspberry Pi acted as a receiver, and another WiFi-connected device — in our case, a laptop — acted as a transmitter. We set the transmitter in a location and then recorded RSSI values using the Raspberry Pi at grid locations in the room. The receiver used a python script referenced as algorithm 1 here. The setup itself is shown in figure 3.

Algorithm 1 RSSI Script

```
import subprocess
import time
import csv
from datetime import datetime

# Configuration
interface = "wlan0" # Change to your Wi-Fi interface name
log_interval = 5 # Seconds between measurements
log_duration = 300 # Total duration in seconds (e.g., 5 minutes)
output_file = "rssi_log.csv"

def get_rssi():
    try:
        r = subprocess.check_output(
            ["iwconfig", interface], stderr=subprocess.STDOUT)
        result = r.decode()
        for line in result.split("\n"):
            if "Signal level=" in line:
                parts = line.split("Signal level=")[1]
                rssi = parts.split("_")[0].replace("dBm", "")
                return int(rssi)
    except Exception as e:
        print(f"Error reading RSSI: {e}")
    return None

# Logging loop
with open(output_file, mode="w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["Timestamp", "RSSI (dBm)"])

    start_time = time.time()
    while time.time() - start_time < log_duration:
        rssi = get_rssi()
        timestamp = datetime.now().strftime("%Y-%m-%d_%H:%M:%S")
        if rssi is not None:
            print(f"{timestamp}_RSSI: {rssi}_dBm")
            writer.writerow([timestamp, rssi])
        else:
            print(f"{timestamp}_RSSI: N/A")
            writer.writerow([timestamp, "N/A"])
        time.sleep(log_interval)

print(f"\nRSSI logging complete. Data saved to {output_file}")
```

Once this data was collected, we modified the Unity C# code for the NCSA WiFi Visualization. For the heatmap approach, during start up we generated receiver assets at locations within the 3D room model, which correlated with the points at which we measured data in the physical room. To do this, we referenced our grid locations in the physical room and mapped them visually to the 3D

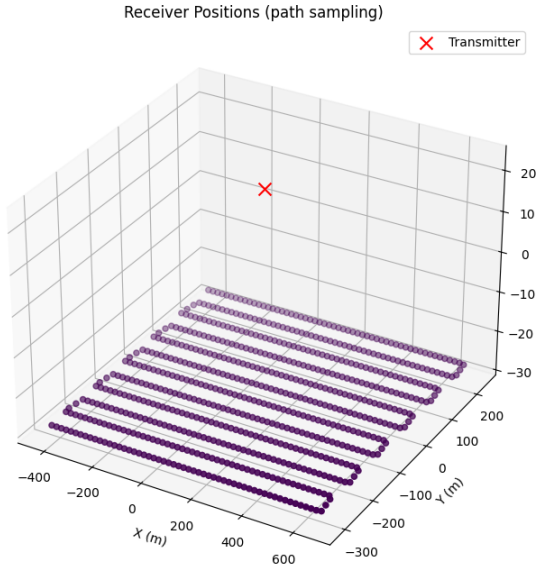


Figure 4: Receiver Placement CSI

model to scale as best as we could. Once the receivers and transmitters were instantiated in the scene, we applied colors to them. The transmitter is blue, while the receivers take on various colors depending on our use case and what values they will represent. For example, for our baseline, the receivers not being tested on took on a black color to indicate to the user these aren't the focus of questioning, while receivers being tested on are gray. For our actual tests and the heatmap itself, all receivers take on a scaled color from green (weak) to red (strong) depending on the RSSI value recorded at that position, using linear interpolation. Any receivers can still be chosen to be grayed out for testing.

This Unity code could also be run on an Oculus Quest 2 headset. However, due to time constraints and shifts in priority, we decided to leave this for possible future work.

Another step that we took was creating a preliminary simulation of the room's CSI values in Google Colab. We loaded the room's .xml file into the Colab notebook. From there we created receivers for the simulation and used SIONNA RT to conduct the ray-tracing on these receivers. The initial setup of the receivers is shown in Figure 4. Once the values of this ray-tracing were created, we stored the CSI values and displayed them in the plot shown in Figure 3. In the future, We will fix the transmitter locations as well as scale the model of the room to fit in Unity. We will also store and display the RSSI values in addition to CSI values.

6 EVALUATION

The setup as described in our implementation and the proposed approach got us user data from 6 randomly chosen users on what they thought the interpolated RSSI values were before and after the heatmap. We have the correct interpolated values and thus we can run analysis and see if their values got closer to the ground truth or farther away after they were given the heatmap.

For our evaluation metrics, we decided to use the mean absolute error (MAE) of the linearly interpolated RSSI values chosen by the users. This metric demonstrates how far off, on average, users' guesses were from the ground truth before and after the heatmap was revealed. Further, we took note of the standard deviation of errors per scenario to possibly demonstrate certainty in each answer among users. Our MAE results are displayed in figure 5.

To help bolster our results, we also asked each user before the heatmap and after the heatmap where they guessed the strongest signal would be, with the correct answer being point B. These figures are displayed in figure 6 and 7, respectively.

From our MAE results, one can see that after the heatmap is shown, the error decreases tremendously. A 72.8%, 34.8%, and 77.3% drop in MAE is seen across points A, B, and C respectively. This is a significant difference, and implies the color scaled receiver heatmap does help users identify relative signal strengths compared to the baseline. It's worth noting that point B, the closest point to the transmitter, dropped its MAE substantially less. We believe this is due to an outlier who had chosen 0.376, compared to the ground truth of 0.778. With a larger sample size, such outliers may not have as large of an effect on the data. Standard deviation also saw drops of 43.7%, 40%, and 83.3% among the three points, which indicated increased confidence among the users. Point C showed a great change in standard deviation, which is an interesting observation given point C has a similar position to point A.

From the PI chart results in figures 6 and 7 we see that most participants were able to initially ascertain that the receiver with the strongest RSSI value was point B. However, there were two outliers, one person switched from point C to B, and the other switched from B to A. In the case of the person who switched from C to B, this is positive as it shows the heatmap helped them identify that point B had the strongest RSSI value. In the other case, this may be attributed to user confusion of the colors and could have lead them to misinterpret the colors on the heatmap. To correct this, we could create a legend to reinforce what colors correlate to what relative RSSI values.

With this data, we can conclude the visual heatmaps do aid users in correctly identifying and determining RSSI values of receivers. This has promising implications for teaching users how to recognize and use a heatmap to identify the places where WiFi signals are strongest.

7 CONCLUSION

Our system of building a heatmap in Unity that displayed RSSI values was highly useful in conducting our user study where we tested if the heatmap aided participants in identifying RSSI values at specific receivers better than without the heatmap. To recap, we asked the users to select interpolated values before and after displaying our heatmap and saw a significant reduction in the Mean absolute error after the heatmap was displayed. This supports the hypothesis that visual heatmaps can aid users in better visualizing WiFi signals in a given room.

We had limitations with the amount of students we were able to interview as it was finals week and people were busy. We were also unable to completely integrate the simulation software that

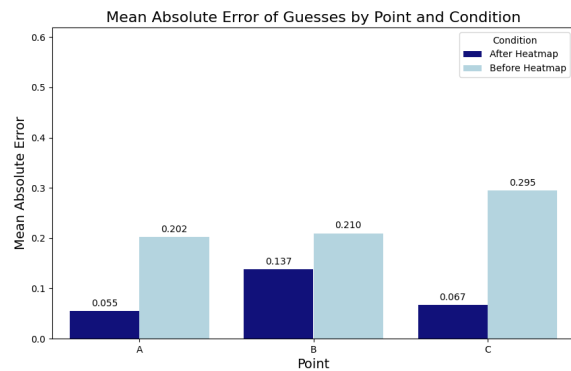


Figure 5: Mean Absolute Error vs Point, for the baseline and after showing the heatmap.

(Before Heatmap) Which point do you think has the strongest signal?
6 responses

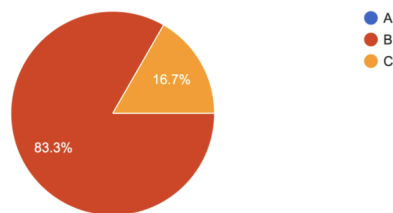


Figure 6: Strongest Point Guess Before Heatmap

(After Heatmap) Which point do you think has the strongest signal?
6 responses

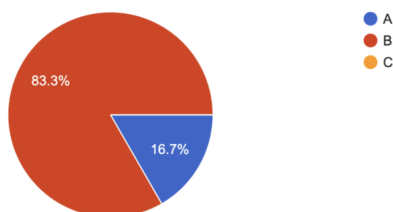


Figure 7: Strongest Point Guess After Heatmap

could automate RSSI collection.

In the future, we can work on using simulations instead of tediously collecting real-world data. We meticulously went around the room with The Raspberry Pi as well as our receiver and recorded these RSSI values which could be done automatically using simulation software. We have the ability to pull CSI values from the room simulation but in the future we would need to scale our model and properly import it into Unity.

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Github: https://github.com/SairamPenumarthy/CS598EKS_Team4/tree/main